



Lean applications across the healthcare ecosystem: Value creation, destruction and a research agenda

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Received: 14 September 2023 / Revised: 20 September 2024 / Accepted: 25 September 2024 / Published online: 8 October 2024
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Abstract

In response to emergent challenges, lean philosophy became one of the most common approaches implemented in healthcare operations. Although lean has been widely implemented, there is still a lack of understanding of the factors that impact value creation in healthcare. Thus, this study aims to identify the factors that impact the implementation and sustainability of lean and value creation in the healthcare ecosystem. We conducted a systematic literature review of 156 articles, and our findings are two-fold: first, we present descriptive results as an outline of the extant literature on lean in healthcare, such as areas of applications, methods, tools, and success and failure cases in different areas of healthcare. Second, we present prescriptive findings and analyse overarching themes ($n=9$) representing significant barriers and enablers of the lean journey. Our study provides implications to the knowledge by offering a multi-level framework that touches staff and stakeholders (people-level), operations and technology (process-level), and organisations and leadership (resources-level). This framework illustrates the impact on value creation/destruction and the need for balanced centricity in the healthcare ecosystem. Finally, we establish a future research agenda with six overarching themes and 18 research questions for scholars and managers to explore the lean journey further.

Keywords Healthcare · Lean · Process improvements · Public healthcare · Value creation · Value destruction

1 Introduction

Replacing the production management systems used until the 1940s, the Toyota vehicle manufacturer developed a methodology that was named ‘Toyota Production System’ (TPS); this system aimed to increase efficiency in the production stages by reducing waste and creating value (Ajmera and Vineet 2019; Vanichchinchai 2023). To understand TPS's philosophy, the Massachusetts Institute of Technology conducted studies on the automobile industry through which TPS was adapted and called Lean Manufacturing between 1985 and 1990 (Womack and Jones 1997). Today, lean is known as a philosophy that is based on five principles:

identifying value from the customer's perspective, mapping the value flow, creating a continuous flow, establishing the organisation of processes in a way that is adapted according to the real need of the customer (pull system); and the search for perfection (Radnor et al. 2012; Antony et al. 2019).

Although lean has its origin in the manufacturing industry, today, this philosophy is adapted and used in different sectors, mainly in services, such as offices, banks, and public institutions, and especially in healthcare services (Tortorella et al. 2021; Fenner and Netland 2023; De Bitencourt et al. 2024). In particular, healthcare services are inserted in a complex ecosystem whereby managers need to balance internal and external demands to reduce costs and, at the same time, adapt the provision of services to the needs of the patients (Waring and Bishop 2010). In response to these problems, lean applications in the healthcare area have enabled a more efficient, patient-centred service with reduced waste (Lindsay et al. 2020; Sales Coll et al. 2024). One of the most remarkable examples of lean in healthcare comes from the Virginia Mason Medical Center (VMMC), Seattle, U.S. According to Kovacevic et al. (2016), the VMMC considers several lean

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improvement tools and techniques, such as rapid improvement events and continuous improvement (kaizen); this approach to lean healthcare ensured VMMC's respectable position in the top 1% of all hospitals in the U.S. in terms of both quality of patient care and system efficiency, with numerous examples of achieved improvements. In this vein, lean has been used with positive results to improve inefficient processes, such as overcrowding, delays, and limited capacity (Volochchuk and Leite 2022; Thakur et al. 2023). The use of lean in healthcare goes beyond emergency areas, and it has been used from basic care and prevention to specialized hospitals that meet the most complex demands, such as hospitalisation, surgeries, procedures, examinations and chemotherapy (Antony et al. 2019).

The literature reports the use of lean in healthcare as beneficial and successful; however, a negative facet of lean sustainability is rarely underlined in the literature (Alkhaldi and Abdallah 2020). Some scholars found that many lean deployments in healthcare are still carried out partially and with difficulties sustaining the results achieved (e.g., Sohal et al. 2021). For instance, a study conducted in outpatient care by Rathi et al. (2021) found little impact on efficiency and productivity after lean implementation; the justification was the staff's lack of knowledge and guidance to conduct and sustain the improvements. The focus on understanding "the dark side of lean" in healthcare is an opportunity to learn from previous experiences and improve the implementation process to avoid unexpected outcomes that will not benefit healthcare operations. Conversely, understanding the enablers of the lean journey provides another opportunity to understand elements that influence value creation in healthcare. The implementation of lean healthcare is a growing phenomenon, and although the literature presents numerous examples of lean implementation, there is still a need for studies that combine and compare elements of the lean journey in healthcare in a holistic way, such as underlying causes of positive and negative results, and the emergence of barriers and facilitators (Kim et al. 2016).

The scarcity of such studies in healthcare might be justified by the complexity of its ecosystem, which has different characteristics from manufacturing, where lean is traditionally implemented. Vargo and Lusch (2016, 11) define a service ecosystem as a 'relatively self-contained, self-adjusting system of resource-integrating actors connected by shared institutional arrangements and mutual value creation through service exchange'. In this study, the healthcare ecosystem considers an inner perspective of the organisation, its sub-areas and actors, excluding a broader external ecosystem that involves providers, manufacturers and other external factors that are considered more difficult to influence and control.

It is in the healthcare ecosystem that patients and staff, as main service actors, collaborate to create value using lean

practices (Womack and Jones 1997). Conversely, value can also deteriorate and lead to destruction; Cui and Osborne (2022) observe that value destruction occurs when service actors fail to integrate resources beneficially. The deterioration of value is an emergent concept that involves users' perception of the service offerings. For instance, the same service that adds value can detract from the lives of service users; this phenomenon occurs when services worsen users' lives (Eriksson et al. 2023). The process of value destruction is also motivated when interactions between service users and providers are diminished, leading to value deterioration, which is understood as the 'dark side of value' (Engen et al. 2021; Leite 2023).

Therefore, because value identification is one of the tenets of the lean culture (e.g., Womack and Jones 1997), studies that analyse the factors that impact the lean journey in the healthcare ecosystem can contribute to increasing the knowledge about value creation/destruction in healthcare (e.g., Lindsay et al. 2020). Furthermore, an emergent concept of value is gaining traction within the healthcare field, which is value-based healthcare (Nilsson et al. 2017). Value-based healthcare is increasingly recognized as a pivotal model for reforming how healthcare services are delivered and financed globally. This approach prioritizes patient outcomes relative to the cost of delivering those outcomes, aiming to enhance the efficiency and effectiveness of healthcare services. According to Porter and Lee (2013), value-based healthcare's core is to reorganize care around patient conditions, improve outcomes, and reduce costs. The concept of value-based resonates to some extent with the perspective of value brought by lean, which is focused on reducing waste and creating value from the users' perspective (what they are willing to pay for). This concept is fundamental in lean and aims to maximise product value while minimizing waste, which is also related to Value-based healthcare, seeks to improve patient health outcomes and make healthcare more cost-effective and patient-centred (Joynt Maddox et al. 2020).

The literature shows that other studies have been investigating lean in healthcare from theoretical and practical lenses, making significant contributions to the literature (e.g. Leite et al. 2024). Nevertheless, a systematic review of the literature that congregates the trends and approaches within the extant lean literature addresses both value creation and destruction in the healthcare ecosystem and sheds light upon the centrality of value in healthcare operations, which is still under-researched. Previous relevant systematic reviews have focused on different elements of lean, such as global perspectives of lean healthcare (Jiju et al. 2019), implementation of lean in healthcare (Lima et al. 2021), and a general perspective of lean as part of process improvements approach (Volochchuk and Leite 2022). Furthermore, the growing search for knowledge about lean in healthcare has motivated several scholars to

suggest new studies in this area, arguing that this can play an essential role in the evolution of lean in the healthcare ecosystem and support healthcare managers to make more informed decisions (e.g., Kam et al. 2021).

Thus, based on the opportunities from the lean literature and the need to support healthcare managers in future implementations and in response to calls for new research, this research aims to identify the factors that impact the implementation and sustainability of lean and value creation in the healthcare ecosystem. Based on this objective and existing literature, we propose the following research questions that will guide our systematic literature review:

RQ1: What is the current outline of the lean literature in healthcare?

RQ2: What factors facilitate and inhibit the lean journey in the healthcare ecosystem?

RQ3: How is lean success and failure in healthcare addressed in the literature?

RQ4: How is value impacted in the healthcare ecosystem?

The study's contributions are three-fold. First, we address the lean healthcare literature presenting the barriers and enablers of the lean journey in a structured format (e.g. Waring and Bishop 2010). Our study moves from solely descriptive barriers and facilitators and analyses them to find overarching themes that represent significant barriers and enablers in healthcare. Second, by addressing this limitation, our results expand the lean healthcare literature to the concept of value creation and destruction in the healthcare ecosystem, tackling the main void in the literature (e.g. Cui and Osborne 2022; Leite et al. 2024). To analyse value in the lean journey, we present a framework that illustrates different influences in the healthcare ecosystem. Moreover, we bring to the fore the importance of the concept of balanced centricity to sustain lean implementation (Verleye et al. 2017). To the best of the authors' knowledge, this is the first study that presents a framework for value creation and destruction in healthcare operations. Third, aiming to provide a descriptive contribution to the literature, we reviewed and organised the lean literature in healthcare. Our findings show the maturity of lean in healthcare, with cases in different countries and areas of healthcare, success and failure cases, as well as methods and tools used to implement lean in healthcare. Finally, the outline of the literature provides a research agenda from the literature with several research questions that contribute to the practice and future studies in lean healthcare. By presenting these three contributions, our study provides a response to research calls in this field of healthcare operations and provides an outline of the current state of lean in healthcare (e.g., Radnor et al. 2012; Lindsay et al. 2020; Leite et al. 2024).

The article is organised as follows: the next section discusses the methodological procedures used in the systematic review of the literature; in sequence, the descriptive and thematic results are presented; the next section discusses the results in the light of literature and its contributions to knowledge and practice; finally, the conclusion summarises the study and presents the limitations and suggestions for future research.

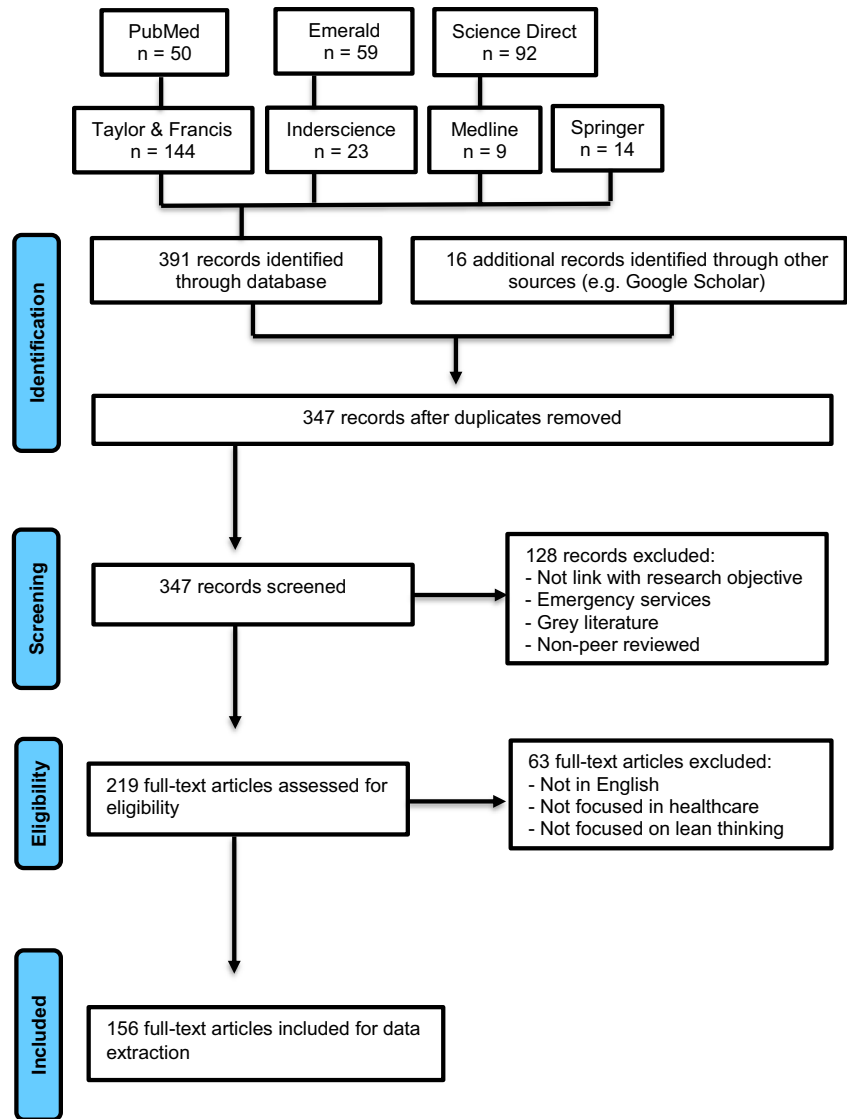
2 Methodology

The systematic literature review (SLR) process is an approach used to extract data from the literature. We used the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol (Volohtchuk and Leite 2022) to conduct and report the SLR, and the procedures were registered in Fig. 1. Additionally, to support a structured report of the data collection, we applied the three phases from Tranfield et al. (2003). The authors propose applying specific principles of the systematic review methodology used in the health sciences linked to managerial research, helping to combat bias and explain the values and assumptions that support a systematic review. When applied with the PRISMA protocol, the three phases improve the quality of the review process, synthesising the research in a systematic, transparent and reproducible way.

In phase 1, initiating the review planning, the specific information and issues that are sought with the study were discussed and defined, such as research objectives, inclusion and exclusion criteria, selection of journals reviewed by specific databases, definition of keywords and research period. The result of these decisions is captured using a review protocol based on PRISMA. The protocol is a plan that helps protect objectivity by explicitly describing the measures to be taken. In addition, the protocol contains information on the specific issues the study addresses, the search strategy for identifying relevant studies, and the inclusion and exclusion criteria (Tranfield et al. 2003).

In the planning phase, it was defined that the review should present the results and lean practices used in healthcare services, which are aligned with the research aim. Because of the maturity of lean in emergency areas and the excess of work published in these areas (e.g., Tlapa et al. 2020; Souza et al. 2021; Tortorella et al. 2021), we decided to focus on other areas of the healthcare sector. For instance, Volohtchuk and Leite (2022) conducted an extensive systematic literature review and identified lean as one of the most used methods of process improvements in emergency areas. The authors reveal that the emergency areas have been one of the first to adopt lean and make suggestions for future studies in different areas of healthcare. In emergency departments, the use of lean is broad, and it has been well-reported

Fig. 1 Systematic Literature Review: PRISMA Flowchart



in literature reviews (e.g., Souza et al. 2021). To illustrate further, Souza et al. (2021) conducted a recent literature review on lean applications in emergency departments. They identified the main barriers and benefits of lean implementation in these settings. Therefore, by excluding the saturated area of emergency and focusing our SLR on different areas of healthcare, our results can avoid unnecessary repetitions and make new contributions, such as in dentistry.

The inclusion criteria of the selected articles were those published in English, with publication date until February 2024, the oldest selected and found article being from 2006, in predetermined databases that presented the keywords and their associations in line with the aim of our SLR. In addition to the selected databases, pilot tests were performed to collect articles in other databases; however, the ones displayed in Fig. 1 were those that returned the studies. The use of the Google Scholar database (additional source) worked

to cross-reference the final information; given that Google has several integrated databases, it was possible to verify if any article was not located in the initial searches performed directly in the original databases.

In the search process, keywords were identified based on articles used in our initial research to identify the research problem and in the literature displayed in the introduction of this article. By doing this, we narrowed the focus of the study to existing studies that underscore a variety of lean experiences in healthcare. We also followed other examples of SLR exploring the impact of lean in healthcare and how they planned their research protocol and defined the keywords to find the highest number of articles published within the literature (i.e., Erthal and Marques 2018; Jiju et al. 2019; Tlapa et al. 2020). Therefore, we selected the keywords 'Healthcare Improvement', 'Lean Healthcare' and 'Dentistry'. One of the aims of SLRs is to identify new

trends and opportunities in the extant literature; hence, the word ‘Dentistry’ was selected to evaluate if there is specific research on the implementation of lean in dental services because this is a topic not so commented in the literature and never mentioned in similar SLRs (i.e. Jiju et al. 2019; Lima et al. 2021; Volochchuk and Leite 2022) which may be a research opportunity (e.g., Robinson et al. 2016). Furthermore, dental services are similar to other operational areas of healthcare that work directly with patients. For instance, dental services deal with an imbalance in patient demand, and it has specific flows and layouts, which sometimes are challenged by “bottlenecks” and increased waiting times. This is different in “low touch” healthcare specialties such as mental health support or nutrition; in these places, the demand is controlled, and the work is focused on cognitive skills rather than operational.

In addition to the keywords in the literature we identified associated words (secondary words) related to the research aims: ‘Process Improvement’, ‘Process Management’, ‘Continue Improvement’, ‘Service Improvement’, ‘Quality Tools’, ‘Practice Improvement’, ‘Sustainability of Healthcare’, and ‘Quality Care’ and ‘Efficiency’.

In the search process, each of the main keywords was combined with each of the associated words, using the Boolean operator ‘AND’. For example, the keyword ‘Lean Healthcare’ was cross-checked with all associated words (e.g. Lean Healthcare AND Service Improvement), and so on. This process of cross-referencing was performed with all the keywords, Booleans and associated words in the databases to ensure that the articles related to the research were located and considered for the study (Table 1). This process was tested in a pilot search in databases, and they returned a high volume of articles. The use of Boolean operators was decided to maximize the combination of keywords and associated words in the SLR protocol. In addition to the “AND” operator, other operators were tested during this phase. The authors tested the “OR” operator (e.g. lean healthcare OR process improvements), and this operator returned very broad results as it also returned articles only about ‘lean healthcare’ or only about ‘process improvements’, for

example. Another test was conducted using the wildcard (*) operator, which represents any combination of characters within a search term, but it did not influence the results obtained during the article collection phase.

As an exclusion criterion, it was decided that would not be used: articles that did not correspond to the objective selected in the research protocol such as those that did not present the word lean, applications of lean in accident and emergency services, conference articles, and those that were not associated with the word lean in management operations (e.g., lean related to anatomy). Furthermore, to access the most relevant secondary data in the literature, we aimed to collect data from peer-reviewed journals, avoiding ‘grey literature’. The search in the selected databases returned 391 articles with an additional 16 articles from Google Scholar, which led to 407 articles found and to be assessed in the following phases of the PRISMA protocol. After the initial selection and removing the duplicated articles, 347 articles were found. After an in-depth analysis with a strict application of exclusion factors, 156 articles were considered eligible for this study. From these articles, we extracted the data that later generated visual aids of the findings.

Following phase 2 proposed by Tranfield et al. (2003) to reduce error and human bias, systematic reviews employ forms of data extraction. To organise the process and systematise the data extraction, a database was created in an MS Excel spreadsheet called ‘data-extraction form’. Tranfield et al. (2003, 216 [emphasis added]) define this form as a spreadsheet that ‘often contain general information (*title, author, publication details*), *study features and specific information (details and methods)* and notes on emerging themes coupled with details of synthesis’. In our extraction form, we considered these suggested details as well as the main lean tools used, the country of the study, main barriers, facilitators, results, and suggestions for future research.

In phase 3, proposed by Tranfield et al. (2003), reports should be produced in two stages. The initial stage is the ‘descriptive analysis’ that should be comprehensive and detailed and provide an overview of the findings in the literature. Later, the ‘thematic analysis’ stage allows the

Table 1 Search structure of keywords, Booleans and associated words

Keywords	Boolean	Associated Words
1st search: Healthcare Improvement 2nd search: Lean Healthcare 3rd search: Dentistry	AND	• Process Improvement • Process Management • Continue Improvement • Service Improvement • Quality Tools • Practice Improvement • Sustainability of Healthcare • Quality Care • Efficiency

researcher to encode and control the data, providing a level of sensitivity to details and context and ensuring accurate access to information. We followed the six steps suggested by Braun and Clarke (2006): (1) familiarizing ourselves with data, (2) constructing initial codes, (3) reading and searching for themes, (4) reviewing themes, (5) defining and naming themes, and (6) producing the report or findings.

Finally, to establish data analysis trustworthiness, we followed similar procedures suggested by Hu et al. (2015) when carrying out SLR, and we formed a review panel to audit the thematic analysis. To form the panel an external member (an academic specialist in thematic analysis) as invited to join the article's authors and audit the codes and themes. When inconsistencies were identified, the panel held meetings to make changes and reach an agreement that would benefit the study.

3 Results

In this section, we present the data found in the literature review using descriptive analysis and later thematic analysis.

3.1 Descriptive analysis

The descriptive analysis presents the research data based on the extraction form and responds to our RQ1 outlining the lean literature in healthcare. The data were transferred to tables and charts to facilitate the understanding of the results.

Figure 2 presents the distribution of publications on the use of Lean by organisations over the years. Published articles were found from 2006 to 2024 (exceptionally, and only for Fig. 2, we excluded the year 2024 because it is the year in progress and in which we collected the data, and the numbers would not reflect the reality of the number of publications in the year). We observed a significant increase

in work after 2013, and it may be the result of the maturity of actions of improvements in healthcare services, making applications in healthcare environments increasingly common. Similar results were found by Volochchuk and Leite (2022) in a recent literature review. The year 2020 had the highest concentration of published studies (17%) which we believe is a reflection of the more mature experience in using lean. In 2021 and 2022, we noticed a reduction in publications, with a slight recovery in 2023, which may have occurred due to the COVID-19 pandemic, which delayed the completion of the articles. The growth of lean studies shows that the healthcare area has attracted considerable attention from academics and professionals with the common goal of implementing lean.

In our survey, we identified the journals that published more articles about lean in healthcare services, which were the International Journal of Health Care Quality Assurance, Journal of Health Organisation and Management, BMC Health Services Research, Leadership in Health Services and The TQM Journal. These journals represent 30% of the publications in the period, according to Table 2, and are journals that focus on quality assurance in healthcare services. The rest of the publications are diluted in more than 70 different journals, without concentrating on the number of publications.

The analysis presented in Fig. 3 shows the geographical distribution where the studies took place, providing information on the location of the countries that most published studies (USA = 28%) on the lean topic in healthcare. The focus that developed countries give to lean is evident, because it is the ones that concentrate 74% of the publications.

Several research methodologies applied to understand the implementation of lean in different healthcare services were also found (Fig. 4). The case study is the most used methodological approach (65%), which is justified by being a common approach to analysing the real context found in the research. For example, in the article by Hung et al. (2015),

Fig. 2 Publication per year

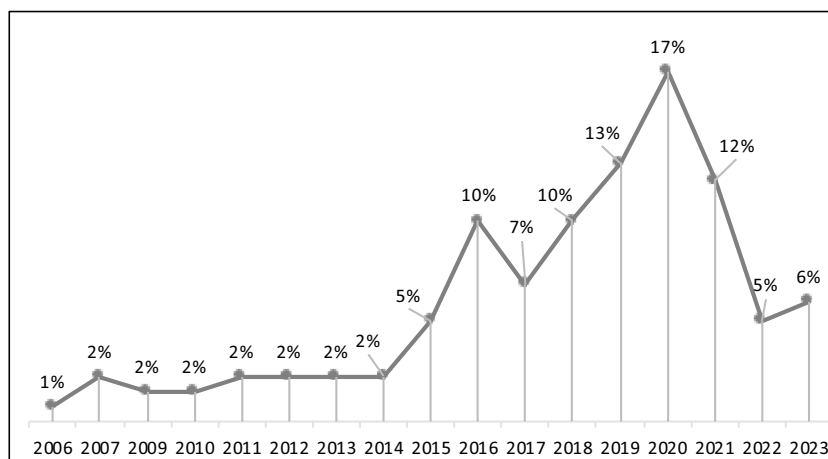
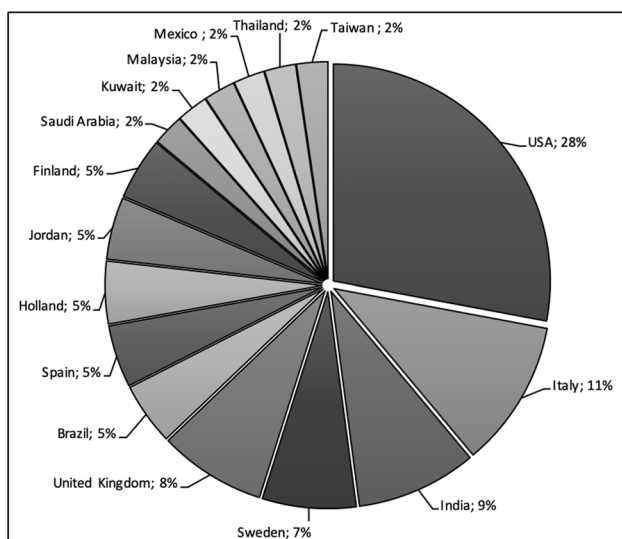
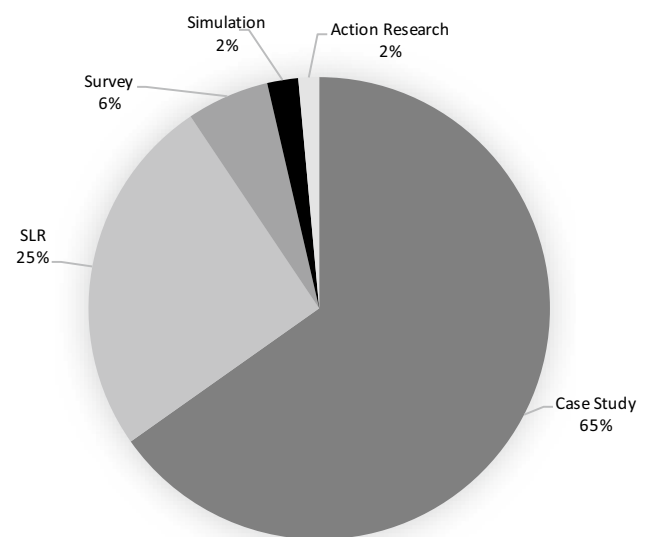


Table 2 Journals with publications in healthcare

Journal	N°	%
International Journal of Health Care Quality Assurance	15	10%
Journal Of Health Organization and Management	8	5%
BMC Health Services Research	8	5%
Leadership In Health Services	8	5%
The TQM Journal	8	5%
International Journal of Lean Six	7	4%
International Journal of Productivity and Performance Management	6	4%
International Journal of Lean Six	5	3%
Business Process Management Journal	4	3%
International Journal of Environmental Research and Public Health	3	2%
International Journal of Quality & Reliability Management	3	2%
Journal Of Evaluation in Clinical Practice	3	2%
Quality Management in Health Care	3	2%
Technological Forecasting & Social Change	3	2%
British Medical Journal	2	1%
International Journal of Health Planning and Management	2	1%
International Journal of Production Economics	2	1%
International Journal of Quality and Service Sciences	2	1%
Brazilian Journal of Operations & Production Management	2	1%
Operations Research Perspectives	2	1%
Acta Ophthalmological	1	1%
Acta Urological Españolas	1	1%
American Journal of Business	1	1%
The Annals of Thoracic Surgery	1	1%
British Journal of Oral And Maxillofacial Surgery	1	1%
Circulation: Cardiovascular Quality and Outcomes	1	1%
Colombian Journal of Anaesthesiology	1	1%
Contemporary Clinical Trials Communications	1	1%
Critical Reviews in Clinical Laboratory Sciences	1	1%
Dyna Management	1	1%
Eastern Mediterranean Health News	1	1%
European Journal of Operational Research	1	1%
European Management Journal	1	1%
Global Health Action	1	1%
Health Care Management	1	1%
Health Policy	1	1%
Health Science Journal	1	1%
Health Services Research	1	1%
Implementation Science Communications	1	1%
Industrial Management & Data Systems	1	1%
International Journal of Health Policy Management	1	1%
International Journal of Intelligent Enterprise	1	1%
International Journal of Lean Enterprise Research	1	1%
International Journal of Six Sigma And Competitive Advantage	1	1%
Interdisciplinary Neurosurgery: Advanced Techniques and Case Management	1	1%
International Journal for Quality in Health Care	1	1%
International Journal of Environmental Research and Public Health	1	1%
International Journal of Health	1	1%
International Journal of Health Planning Management	1	1%
International Journal of Healthcare Management	1	1%
International Journal of Organizational Analysis	1	1%

Table 2 (continued)

Journal	N°	%
International Journal of Production Research	1	1%
Journal of The American College of Radiology	1	1%
Journal of Medical Systems	1	1%
Journal of Nursing Care Quality	1	1%
Journal of Hospital Medicine	1	1%
Journal of Healthcare Engineering	1	1%
Journal of Biosciences and Medicines	1	1%
Journal of Clinical and Diagnostic Research	1	1%
Journal of Paediatric Nursing	1	1%
Journal of Surgical Research	1	1%
Materials Today: Proceedings	1	1%
Nursing Clinics of North America	1	1%
Operations Research For Health Care	1	1%
Oral Health Practice Faculty Publications	1	1%
Orthopaedic Clinics of North America	1	1%
Quality And Safety In Health Care	1	1%
Quality Management In Primary Care	1	1%
Retina, The Journal of Retinal And Vitreous Diseases	1	1%
Revista Brasileira De Enfermagem	1	1%
Revista De Calidad Asistencial	1	1%
Revista Gaucha De Enfermagem	1	1%
Sustainability	1	1%
Systematic Reviews	1	1%
Total Quality Management & Business Excellence	1	1%
Tropical Medicine and Health	1	1%
Value In Health	1	1%
The Journal of Evidence and Quality In Health Care	1	1%
Journal of Management Control	1	1%
Production Planning & Control	1	1%

**Fig. 3** Countries with publications**Fig. 4** Methodologies used

there is the report of a case study on the implementation of lean in an outpatient service which obtained positive results, such as employee engagement and improvement of management performance. The second most common research method was the systematic literature review (SLR), appearing in 25% of the articles, showing the importance of providing a theoretical basis. All SLR articles found positive results in the implementation of lean, except for an article that reported negative results, which shows the team's discontent with the implementation of lean (Gowen et al. 2012). Other methods, such as surveys, simulation, and action research, focus on 10% of the articles.

The application of lean in different healthcare services (excluding emergency areas) is shown in Fig. 5. From the selected studies, 39% occurred in general hospitals, where, for example, a 94% reduction was found by Bhat et al. (2014) in the waiting time of patients in the queue for elective surgical procedures. Among the outpatient clinics, 19% of the articles present the implementation of lean in different specialties. For example, in ophthalmology, Kam et al. (2021) reported that the redesign of the patient flow adjusted and met the demands and tasks performed by the teams, optimizing the scheduling and reducing the waiting time. According to Hung et al. (2016), the improvements occurred in all processes and teamwork proved to be efficient to find positive results in the implementation. Dental services have few published studies on lean implementation; only 2% of the selected studies were carried out in these places, demonstrating an opportunity for research in this area. Regardless of the location, lean implementation is useful in the search for different purposes, such as streamlining and redesigning the work processes and examining the effectiveness of the current processes. This shows that lean is no longer restricted to emergency areas.

In order to understand the technical details of the implementation of lean (i.e., Vanichchinchai 2023), we conducted

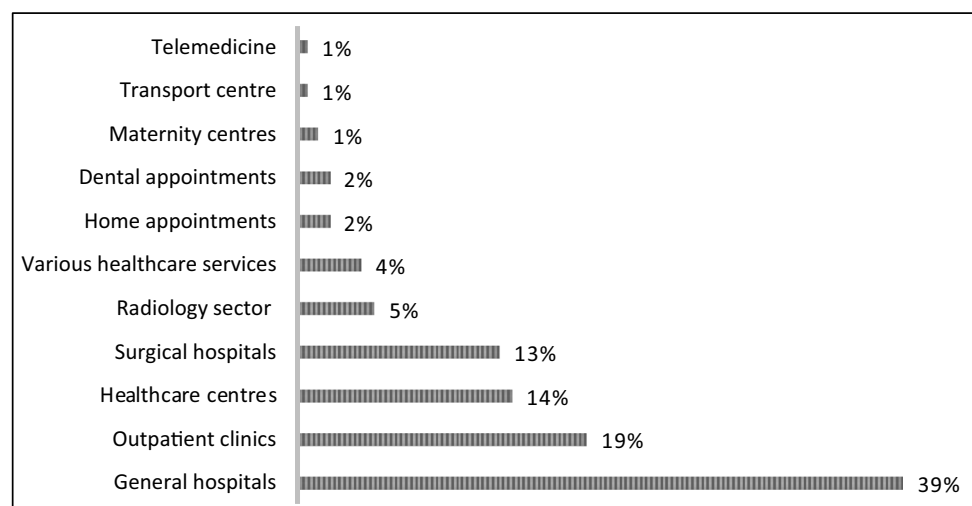
a survey of which lean tools were the most used (Fig. 6). Value Stream Mapping was found in 22% of the articles, demonstrating the importance of measuring the processes (e.g., Ahmad et al. 2022). According to Poksinska et al. (2017), an important strategy of lean is to develop systematic approaches to define value that integrates both the knowledge of the healthcare team and the preferences and needs of patients. Mapping processes leads the team to reflect on their actions, their times and how the work process is being carried out. According to Verma and Amar (2015), this analysis defines and requires the optimisation of the value for care (where the value is defined in better care, cost and health), with the need for new approaches for patient care. Other tools are constantly used, and Gowen et al. (2012) observed a study in which DMAIC (Lean Six Sigma) and 5S were used and achieved a decrease in patient waiting time, cost reduction and hospital redesign for patient-centred care.

In line with RQ1, this descriptive analysis demonstrates the diversification and growth of the use of lean in different areas. The results show the importance of using lean tools for the implementation and sustainability of this methodology in different healthcare services, as well as the need for future research that demonstrates its relevance. Following this, we present an in-depth analysis of the barriers and facilitators of the implementation of lean.

3.2 Thematic analysis

Thematic analysis is a common form of presentation of qualitative analysis that identifies relevant themes that arise from the codes found in the articles (Braun and Clarke 2006). Thus, aligned with RQ2, we seek to select the elements that serve as barriers and enablers in the lean journey. Once the process of thematic analysis was initiated, it was necessary to search for common codes related to the facilitators of the

Fig. 5 Areas with lean applications



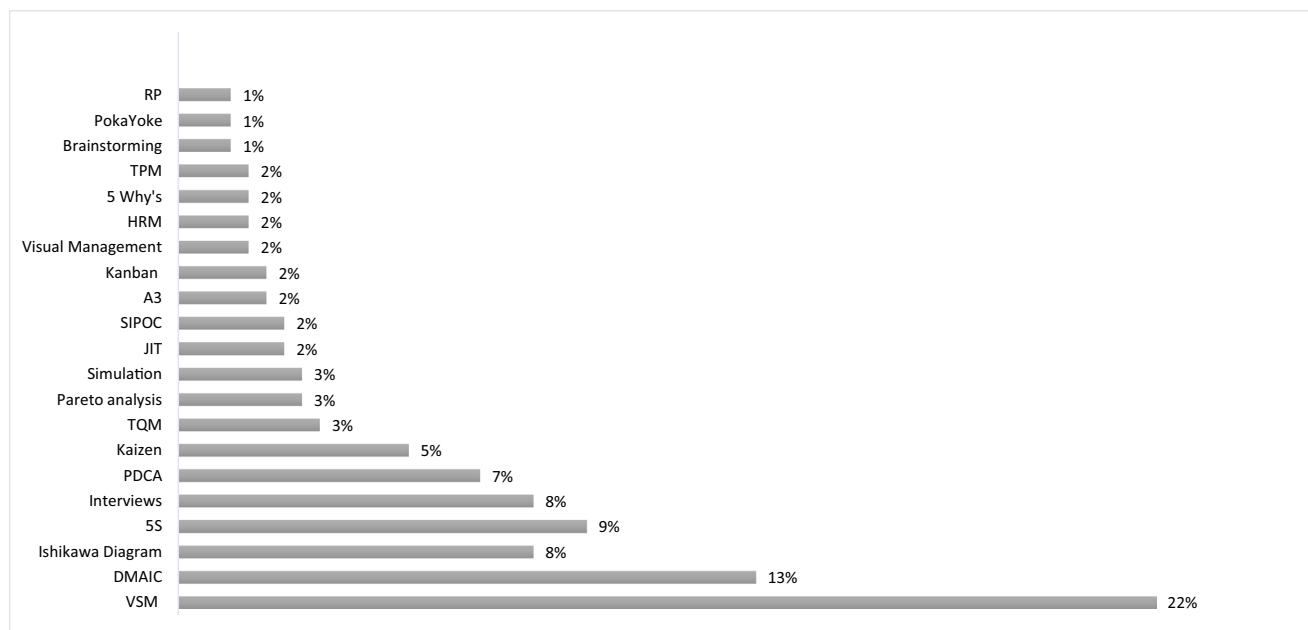


Fig. 6 Tools used in the implementation of lean

process, that is, extract and encode the inherent elements that facilitated or hindered the implementation of lean. The objective is to remove the codes from the articles group them according to the frequency of repetition in the literature, and then organise these codes by themes. The themes, according to Braun and Clarke (2006), are defined by an identification

process, providing specific names and aspects, leading the researcher to reach a degree of consensus on the themes presented. Table 3 shows the codes that led to the creation of the themes that are enablers for the application of lean. These codes demonstrate the ostensible elements that motivate enabling themes in the implementation process.

Table 3 Themes and codes: enablers

Themes	Percentage of times mentioned in literature	Enablers / Codes
Leadership commitment in the implementation	27%	• Leaders involved • Management support • Clear communication
Lean as an organisational strategy	25%	• Responsibility for the sustainability of the process • Waste reduction • Improvements in work processes • Lean philosophy in the organisational culture • Organisational focus on patient satisfaction • Use of improvement tools
Team involvement	24%	• Engaged team • Team training
Planning for Lean implementation	24%	• Evidence-based decisions • Performance indicators • Aspiration for new solutions • Identification of problems • Ability to deploy new technologies • Focus on medium and long-term implementation • Areas pre-defined for deployment

The first facilitator theme of lean is ‘leadership commitment’, revealing the importance of effective leadership and governance as one of the main success factors in the implementation. According to Deara et al. (2018), it is essential for management to want to implement lean. The commitment of leadership within the process makes a difference in results because a leader involved and who communicates clearly with the team, acts as a sustainability factor in the implementation of lean.

The next topic deals with ‘lean as an organisational strategy’ in services. The organisation that aims to implement the lean model must have lean thinking evidenced in its mission and values. Lean needs to be in the culture of the organisation; according to Kim et al. (2006) lean provides a management philosophy and tools that lead to transformation. This transformation should seek the elimination of waste and the improvement of work processes, focusing on patient satisfaction.

As a third facilitator theme ‘team involvement’ emerged in the implementation of Lean. Engagement with lean training provides employees with improved performance, incorporating a culture of innovation, increasing commitment and support in implementation, and generating a more positive attitude toward the proposed changes (Hung et al. 2015). It is essential that teams understand and commit to the implementation process; real involvement is necessary, that is, that the whole process of change makes sense in the practice of their daily routines.

In sequence, we identified the need for ‘planning the implementation of Lean’ as an enabler. The institution must be clear about the problem it wishes to address and the goals and results it wishes to achieve, and must have some form of monitoring and evaluation (Ajmera and Vineet 2019). Planning should also take into account the selection of the site and the most appropriate time for the implementation of lean. According to Prado-Prado et al. (2020), the choice of the site can guarantee the success of the implantation, being essential to start with areas that facilitate the implantation and identify the existing structural conditions. In that sense, starting with sectors or departments not only guarantees greater success but requires the use of fewer resources and the use of simpler tools.

The description of the themes that facilitate the implementation of lean, arises as support for managers who wish to implement lean. According to Rossum et al. (2016), many healthcare organisations encounter difficulties when the implementation of change initiatives presents a gap between strategy and implementation, which is when enablers can help to reduce this problem. In order to present more evidence from the literature for each topic found, we created Appendix I.

Finally, we identified the barriers to the implementation of lean. In Table 4, there are codes that function as barriers in the implementation of lean, which demonstrate the elements inherent to each theme.

Table 4 Themes and codes: barriers

Themes	Percentage of times mentioned in literature	Barriers / Codes
Inadequate understanding of Lean thinking	34%	• Continuous improvement processes without sustainability • Absence of Lean thinking in organisational culture • Lack of knowledge about Lean methodology • Limited knowledge in the use of improvement tools • Waste of workers' skills • Indifference toward the opinion of the patient
Insufficient preparation for implementation	25%	• Dubious reliability of data • Imbalance between capacity and demand • Scarcity of financial resources • Immediacy in the implementation • Inadequate infrastructure
Resistance to change	21%	• Reluctance of the teams • Antagonism of the management team
Lack of leadership	12%	• Omission of the leadership in promoting change • Poor communication • Inexperience in deployment
Poor work processes	8%	• Fragility in existing flows • Absence of training • Existence of waste

The first topic that appears in most articles as a factor that hinders lean implementation is the ‘inadequate understanding of lean thinking’. For most healthcare organisations, knowledge of lean thinking represents a barrier, both in the implementation of lean and in the maintenance of improvements. The lack of knowledge about lean by healthcare teams makes it necessary to overcome the arguments about the application of lean manufacturing concepts in the healthcare sector, with the claims that ‘people are not automobiles’ and ‘each patient is unique’ as Kim et al. (2006) reported. If the organisation does not have in its culture that improvement processes must be fundamental and with a focus on patient satisfaction, it is difficult to sustain the implantation. In view of these situations, according to Rossum et al. (2016), there is a dependence on external support in healthcare services, which also hinders the sustainability of the implementation.

The next barrier found is ‘insufficient preparation for lean implementation’. The lack of some previous requirements, such as data reliability, a large imbalance between installed capacity and demand, inadequate infrastructure, lack of standardised work and indicators, and lack of financial resources, demonstrate that the institution is not adequately prepared, and the implementation may be impaired. According to Bowerman et al. (2007), planning is essential for successful implementation because lean will not solve all the problems of the institution, but if there is an adequate deployment project, it will be easier to achieve the expected success. Still associated with insufficient planning an extremely important factor is expectation for immediate results. For Bhat et al. (2014), impatience in achieving results also leads to failure. Prior preparation with a good analysis of existing data and requirements can prevent the wear of a frustrated implementation.

‘Resistance to change’ appears as a barrier during lean deployment (Mishra and Sharma 2024). One of the justifications found, both in team members and in leaders, for resistance and negative attitudes is the fear of changes and the possibility of layoffs. Erthal et al. (2021) also say that the literature identifies a defensive mechanism as ‘regression’ when the old solutions and security of the past are desired. Also, teams do not want to expose themselves and are afraid of punishment, so they cling to old and safe solutions, which characterize a ‘regression’. In addition to the fear of change through these factors, the proposal to review the work processes and the search for standardisation and division of well-defined tasks causes distrust. The team, and especially doctors, feel insecure fearing the loss of autonomy, so they resist and hinder the implantation. In addition to looking at the individual work process, teams need to understand that by working together and in an organised way, the answers will be more appropriate and more likely to succeed (Hung et al. 2015; Narayanamurthy et al. 2018; Erthal et al. 2021).

Another barrier found was the ‘lack of leadership’ during the implementation of lean. For the leadership to be present and sustain a process of change, it must know the implementation process and its tools (Poksinska 2014). Healthcare leaders and managers often do not know lean, are inexperienced in the implementation process and end up remaining silent in promoting the necessary changes or not communicating the proposals efficiently. Addressing the issue of effective communication by the leadership, Deara et al. (2018) observed that one of the biggest barriers found in the use of lean is the use of new terminologies and, consequently, the difficulty of communicating lean principles (e.g. Lean Lexicon).

The ‘poor work processes’ are presented as a theme that collaborates for the formation of a barrier during the implementation. An institution that does not manage its information, such as collection, storage, analysis, monitoring and control of data, will have difficulties in the implementation and maintenance of lean (Prado-Prado et al. 2020). Along with the lack of data management, Lima et al. (2021) identify that lack of team training, lack of conceptual alignment and formal training are processes that can create obstacles to achieving sustainable lean implementations in the long term. Other difficulties related to deficient processes are the existence of waste and the fragility of established flows. As exemplified by Kokshagina (2021) the patient follow-up processes, information and feedback are collected only if complaints exist and not in a systematic way. In Appendix II, we present excerpts from some articles that indicate evidence from the literature for each topic found.

After presenting the themes, the importance of knowing both the themes that facilitate the implementation of lean and those that present themselves as barriers that affect the sustainability of lean and its ostensible elements is evident. Furthermore, the thematic analysis presents some overlaps between themes from barriers and enablers (e.g., Lack of leadership versus Leadership commitment in the implementation or Insufficient preparation for implementation versus Planning for lean implementation); these themes overlap and act as countermeasures for sustainable lean implementation.

3.3 Lean applications in practice

Lean is not a ‘silver bullet’ that solves all the problems of healthcare operations; in fact, many of the lean projects end up not delivering results; however, this is rarely addressed in the literature. Hence, in this section, our research addresses lean in practice, and shows both sides of the lean journey (e.g. positive and negative results), and sheds light upon our RQ2.

Although most of the articles published on lean focus on the positive results of the implementation, we also believe

Table 5 Negative Results of Lean Application

Source	Area of healthcare	Summary of the problem	Result achieved	Justifications for negative results
Xie and Peng (2012)	Hospital	Reduce patient waiting time and increase use of surgical centres	There were no improvements in some patient care times	Bottlenecks in patient flows and lack of priorities from the organisation
Gowen et al. (2012)	Outpatient clinic	The rapid increase in healthcare costs, partly due to preventable medical errors, has led to the need to improve processes	The results of quality improvement were only moderate, failed and lacked team commitment	The inadequate choice of process improvement programs and the lack of acceptance of stakeholders
Poksinska et al. (2017)	Healthcare centre	Little evidence that lean in healthcare should contribute to improving patient satisfaction	Little attention given to definition and value from the perspective of the patient	No tools and methods were found to improve customer participation in the value creation process
Hung et al. (2018)	Healthcare centre Outpatient treatment	Need for innovations in the workforce, including redesign of teams	Team discontent during deployment	The team presented higher levels of burnout and workplace perceptions as stressful. The increase in stress reports was higher among doctors
Kelendar et al. (2020)	Healthcare centre	Need to increase the care of chronic patients by healthcare systems	Lean presented inconsistent benefits, in the results of processes and in the patient flow. Little or no impact on process efficiency	The superficial participation of patients and the technical application had a limited organisational scope
Alkhalidi and Abdallah (2020)	Hospital	Need strategies for healthcare companies to succeed by combining better and more efficient operations, reducing costs, and reducing waste	Improvements have led to higher performance in quality, but not efficiency or accessibility	The individuality of patients makes quality of care more important than efficiency or accessibility in terms of techniques
Tlapa et al. (2020)	Hospital	Reduce waiting times for patients in care	Lack of evidence of the impact and benefits of lean on patient/employee satisfaction	Low patients and employees participation
Erthal et al. (2021)	Hospital	Cultural resistance that naturally emerges during lean implementations	Team discontent during deployment	The non-effective management of interrelated and complementary factors that are: learning, organisation, achievement and performance impaired the results
Rathi et al. (2021)	Outpatient Care	Improve efficiency and productivity of the healthcare services	Little impact on efficiency and productivity	Healthcare professionals lacked guidance on how to create a culture of continuous improvement and how to sustain improved processes
Reponen et al. (2021)	Hospital	Low performance in healthcare operations	Little impact on efficiency and productivity	There is no consensus on the dimensions of performance measurement and benchmarking in the lean healthcare

it is important to acknowledge negative results as learning experiences. It is possible to understand the situations that demonstrate when lean did not work and for what reasons it occurred. In addition, these reasons create conditions to avoid difficulties in deployment and thus increase the possibility of success. Table 5 details the operational problems, the results and the conclusions found, in addition to the sources and location of the research.

In most articles that presented negative results, the problems that lead to the implementation of lean end up being similar, among them the need to improve processes and reduce costs. The results identified a negative focus with little or no impact on improving efficiency and team discontent.

Negative results were identified in different areas of healthcare sectors, such as outpatient clinics, healthcare centres and hospitals. They are justified by failures in the implementation, from the lack of management support to the lack of acceptance by the teams and the absence of a consensus in the measurement of lean performance (e.g., Erthal et al. 2021; Reponen et al. 2021). Still, as a negative result, little attention is given to the participation of the patient in the process of change, justifying the lack of identification of tools and methods to improve the participation of the patient (e.g., Tlapa et al. 2020). Performing a cross-reference analysis with the factors that facilitate the implementation of lean (Table 3), many of the justifications and conclusions found in the negative results could have been avoided.

In contrast to the negative results, Table 6 shows the positive results of lean implementation. All selected articles presented an operational problem, the need for improvement, either in access, waiting time or in the quality of the services provided. Most studies have found positive results for these problems, identifying that the implementation of lean resulted in increased efficiency, cost reduction, optimisation and reorganisation of processes. Positive results were also found, like the improvement in the patient's experience during their care and increased satisfaction, quality and safety in the care received.

Observing the positive results found from the perspective of healthcare organisations, one of the contributions presented is the improvement of cost/efficiency, making the service more economically viable (e.g., Narayanamurthy et al. 2018). For workers, positive contributions appear as the motivation found to promote changes, the identification of waste and the search for quality for the patient (e.g., Simons et al. 2017). Moreover, positive results were also perceived by the patients, for instance, a service with greater safety and quality, with better access and at the right time (e.g., Wang et al. 2015; Robinson et al. 2016).

4 Discussion

Our results have shown how lean is implemented and sustained in healthcare settings. These results echo one of the most germane lean tenets, value identification (e.g., Womack and Jones 1997; Ahmad et al. 2022). According to Radnor et al. (2012), the definition of value from the user perspective is what drives process improvement initiatives. For Osborne et al. (2022), the creation of value can be shared with service providers (co-creation), but the user is the ultimate value arbiter. In the same vein, Cui and Osborne (2022) acknowledge that the user can define when value is deteriorating and leading to value destruction. In our research, barriers, and enablers have led to failure and success in lean implementation in healthcare, bringing the importance of value identification to the fore. Thus, under the lens of value creation and destruction, and aiming to answer RQ4, we discuss the results of this study and provide contributions to theory and practice.

4.1 Contributions to theory

To understand the complexity of the healthcare ecosystem and how value creation and destruction are affected, we adopted a configuration of the service ecosystem suggested by Maglio et al. (2009) to create a multi-level healthcare framework. According to the authors, a configuration of people, technology, processes, and resources represents an abstraction of the service ecosystem for the twenty-first century. In that sense, the healthcare sector can be viewed as a service ecosystem in which interactions within and across the levels of the ecosystem impact mutual value. The original framework from Maglio et al. (2009) does not explicitly underscore 'process' in its configuration; instead, it identifies technology as part of the process, but the authors suggest that the framework can be adapted to different contexts, such as healthcare. Thus, we redefined this level as a 'process', which better resonates with our findings and discussions (Fig. 7).

The healthcare ecosystem reflects the interplay between different levels of the framework in the lean journey and how they impact or are impacted by the barriers and enablers, resulting in value creation or destruction. The left-hand side of the framework is where the enablers of the lean journey are placed; as these are facilitators of the lean journey, they are affected by the levels of the healthcare ecosystem. Conversely, on the right-hand side, the barriers are displayed as they work as barriers to lean implementation; they affect the central levels. Most of these barriers and enablers overlap and could be related to different levels of the framework; however, for the purpose of this discussion and to keep it focused, we connected the two dominant barriers and enablers with the most related level.

Table 6 Positive Results of the Lean Application

Source	Area of healthcare	Summary of the problem	Results achieved	Contributions of the Study
Taner et al. (2011)	Radiology clinic	Failure in the image department	Improved workflow by eliminating the causes of failure. Impact on the cost and quality of services	Improvements in the study workflow, made by determining the failures and potential risks associated with radiological care, had a positive impact on patient safety. Having eliminated repeated examinations, the risk of being exposed to more radiation has also been minimized
Wang et al. (2015)	Healthcare centre	Long waiting times in healthcare services	The waiting time was reduced by about 56%	Reducing the waiting time and investigating the size of the most economically viable customer group is useful to resolve time and cost issues in healthcare processes
Kanamori et al. (2016)	Outpatient clinic	Improve the quality of government healthcare services	The 5S program brought a culture of continuous learning and care	The use of 5S improves the quality of healthcare services and employee motivation
Andersson and Hilletoft (2015)	Home care	Challenges of the healthcare processes in the geriatric care sector	Some processes have been reorganised to be more efficient, improving the workflow	Lean implementation leads employees to realise there is a large amount of waste in the processes
Robinson et al. (2016)	Dental clinic	Waiting time of more than four hours for the appointment	One-hour reduction in the waiting time, improved patient satisfaction by 21% and reduced negative comments by 24%	The improvement process resulted in shorter treatment times, greater patient satisfaction and improved collaboration. It suggests that the lean process has applications in healthcare environments, including dental clinics
Paim et al. (2016)	Maternity	Poor schedule system to manage the demand for obstetric surgeries	A 70% reduction in the cancellations, an increase in the number of scheduled procedures in the units, and an increase of 97% in the level of scheduling appointments	Persistence in believing and trusting a new cultural paradigm of thinking about healthcare projects, in a structured, planned and coordinated form. Delivering excellence in care
Simons et al. (2017)	Radiotherapy	Delays in the treatment of oncology	Organisational improvement improved the waiting time, patient safety, employee satisfaction and short-term absenteeism	Continuous evaluation of effects to study sustainability of long-term improvements
Chang et al. (2018)	Surgical Centre	Need for improvement of time and efficiency in surgical rooms	The results showed that a standardised surgical process can reduce the total waiting time	The discovery presents an understandable model that covers research work and a holistic view of improving patient satisfaction and healthcare quality
Narayanamurthy et al. (2018)	Hospital	Reduce the cost of care in healthcare institutions	The benefits of the deployment bring competitive advantage by providing quality treatment at a low cost in the market	The training sequence of employees at different levels in the hospital can help the manager in the implementation of Lean

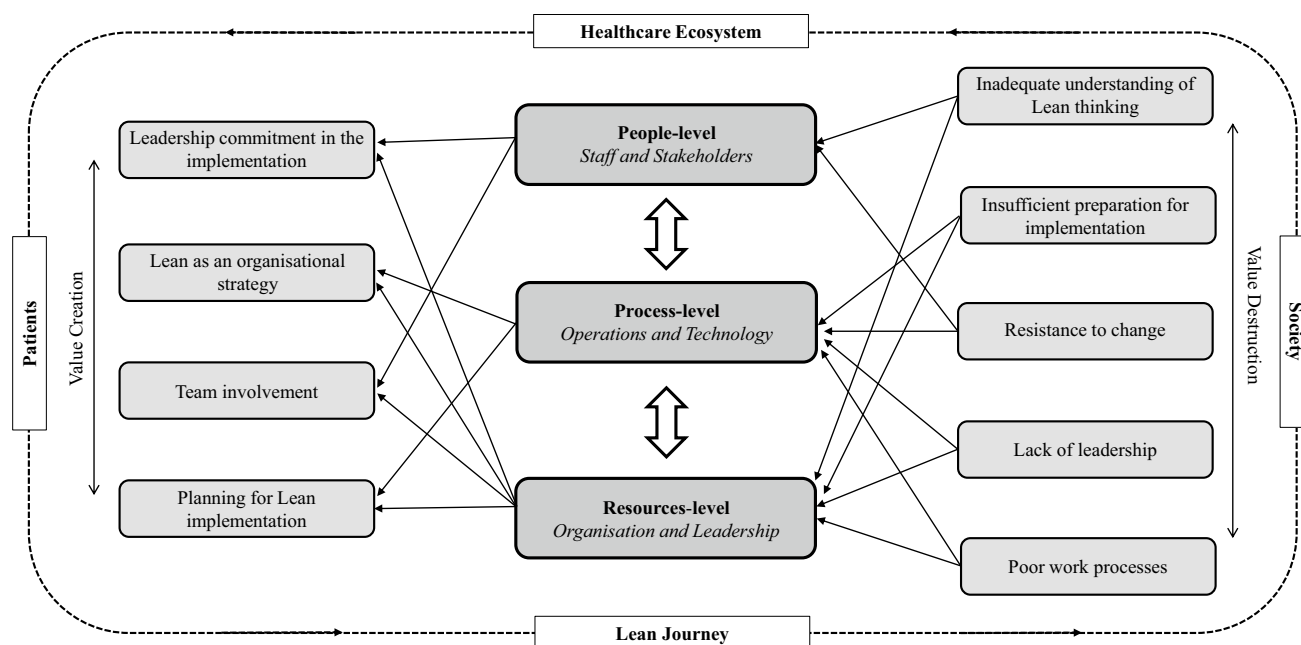


Fig. 7 The Healthcare Ecosystem Framework

The ‘people-level’ represents healthcare staff and key stakeholders; in this level, barriers such as the lack of understanding of lean and the resistance to change represent significant barriers to the lean journey, impacting people involved in the implementation process. The literature presents results related to the impact of these barriers on people; Waring and Bishop (2010) identified lines of resistance within the healthcare staff, representing prominent obstacles not just for lean but also for future re-configuration of clinical practices (e.g. new protocols and technology implementation). In response to these barriers, the framework presents enablers of the lean journey that are influenced by the people-level. For instance, leadership commitment and team involvement in the implementation are crucial facilitators in healthcare. The active involvement of people in the process has been suggested by Radcliffe et al. (2020) as an essential enabler to increase staff engagement in process improvement programmes, which is known to enhance team experience and commitment. This echoes findings from Leite et al. (2024), who suggest that for a sustainable implementation of lean, healthcare managers need to involve key stakeholders to co-design the healthcare process.

The impact on healthcare operations and technology is illustrated by the ‘process-level’; for Maglio et al. (2009), this level of the framework interacts with other parts of the service ecosystem to create mutual value. Our study identified several barriers at this level; however, barriers such as inadequate preparation and poor work processes considerably impact healthcare operations, processes and patient well-being. For instance, Mahmoud et al. (2021) advocate that financial investments per se have limited impact on

improvements; however, if inefficient processes are not addressed at the beginning of the project, they will lead to difficulties in implementing lean. In response to these barriers, the process-level might facilitate the creation of lean enablers, such as lean as an organisation strategy and planning for lean implementation. A recent work by Ajmera and Vineet (2019) underlines some success factors, such as leadership, communication and training, as the bedrock for lean implementation and sustainability. Finally, the literature shows that addressing organisational culture, such as poor strategy and vision when implementing lean, might lead to positive results and avoid barriers at the process-level.

The last level is labelled as ‘resources’, and Maglio et al. (2009) provide several definitions for resources but conclude that organisations are conceptual resources with legal rights, competencies, skills and knowledge. In that sense, the healthcare framework places the resources-level as the representation of organisation and leadership. In the framework, this is the only level related to all barriers and enablers identified in our SLR, underlining the importance of organisation and leadership direction for a sustainable lean implementation (Radnor et al. 2012). The emergence of barriers, such as lack of leadership and inadequate understanding of lean thinking, affect this framework level. For instance, Radcliffe et al. (2020) observed that some participants in the hospital still associate lean with the manufacturing context, posing difficulties in inserting lean in the healthcare environment. This is a common mistake where organisations implement lean as a ‘one size fits all’ strategy; however, lean is context-dependent and must be adapted to achieve positive results

(Williams and Radnor 2018; Leite et al. 2020). The framework presents enablers to counter these barriers, such as leadership commitment in the implementation and lean as an organisational strategy (Kim et al. 2006). The former highlights the importance of leadership in the changing process, whereby Waring and Bishop (2010) suggest that healthcare reforms rely on how leaders persuade clinicians to change the process, focusing on improvements and customer value.

The identification and preservation of value are at the heart of lean culture (e.g., Womack and Jones 1997). The healthcare framework presents two different facets of value: creation and destruction. The latter represent imbalances in the healthcare ecosystem, characterised by barriers that inhibit value creation for the patients and society. Cui and Osborne (2022) explain this phenomenon by describing the ecosystem as a complex network where service actors interact. However, value destruction occurs when these actors fail to integrate resources beneficially.

To some extent, the five barriers identified in our study represent potential value deterioration and failures in the healthcare ecosystem (Table 5); this was also identified in the literature as value contamination and the ‘dark side’ of value by Williams, Kang and Johnson (2016). Conversely, value creation is characterised by the enablers of lean implementation, and successful projects represent more comprehensive benefits that affect patients and society alike (Table 6). In the framework, the three levels of the service ecosystem interact with the value from different perspectives; however, the patient and society as users define the value status (e.g. value arbiters). According to Osborne (2018), service organisations only establish value propositions (service offerings) and facilitate value creation through users’ interactions and experiences with the service. Thus, the three levels support value creation within the healthcare ecosystem by creating enablers for lean implementation and sustainability.

Finally, to achieve value creation and avoid destruction in the healthcare ecosystem, it is essential to secure the interest of all stakeholders involved in the process; Verleye et al. (2017) report that when at least one actor in the service ecosystem does not have their interest secured, this might result in imbalance and value deterioration. This echoes the concept of balanced centrality in the service ecosystem, defined by Gummeson (2007) as the need to secure the interests of multiple actors, including the user as a beneficiary. In the healthcare sector, Leite et al. (2024) have suggested the introduction of a balanced centrality concept to ease the impact of emergent barriers during lean implementation. Therefore, we believe that finding the balance between different actors’ interests (e.g. patients, physicians, nurses, organisations and society) in the healthcare ecosystem might increase the chances of a successful implementation. For instance, our results show several examples of successful and unsuccessful lean projects (Tables 5 and 6);

hence, lessons learned from these organisations might help healthcare managers and lean practitioners avoid barriers and create enablers in healthcare settings.

Value destruction is an emergent concept in lean literature; thus far, the literature has only addressed non-value activities such as waste (Nilsson et al. 2017; Mishra and Sharma 2024). By bringing the concept of value destruction, we expand the lean literature, explaining the impact created when a lean process makes the life of the user worse (Cui and Osborne 2022). A faulty or ‘broken’ process is not only creating waste from the organization’s perspective, but it is also detracting value from the service user, leading to the destruction of the value identified. Furthermore, our research shows that value destruction and creation are on the opposite side of a continuum, where enablers are facilitating the creation of value, and barriers are contributing to value deterioration. Finally, looking into the healthcare ecosystem framework, we find that the interplay between the creation and destruction of value is based on a ‘fine line’ where, depending on how people, processes, and resources interact in the healthcare ecosystem, it can easily turn value creation into value destruction. The framework helps to focus on core organizational areas that can lay the foundation of a stable ecosystem based on value creation.

4.2 Implications to practitioners and researchers

Our results make broader contributions to practice by shedding light upon the challenges of lean implementation and the current ‘status’ of lean in the healthcare sector (Figs. 2, 3, and 4). Overall, the study provides a foundational framework for advancing both the practice and study of lean in healthcare. It not only addresses current knowledge gaps but also sets the stage for future research that can contribute to the evolution of lean methodologies tailored to healthcare’s unique challenges and opportunities. Hence, we contribute to healthcare managers, lean practitioners, and researchers by providing examples of lean in different healthcare areas (Fig. 5), different tools (Fig. 6), and how to use the enablers to counter barriers that emerge during the lean journey (Fig. 7). Despite some negative results and piecemeal implementations, lean has been positively used in healthcare, and our study highlights its benefits. Therefore, we suggest that healthcare organisations and policy-makers develop regulations to adopt lean as a common practice to improve public and private healthcare services for the benefit of patients and society.

Our study aims to expand the contributions to scholars and healthcare managers. Therefore, drawn from our findings in the literature, we outline a research agenda with research questions that are expected to motivate other researchers and healthcare practitioners to address timely issues (Table 7). The table was created based on the SLR analysis of the ‘research calls’ theme repetition. For instance, the percentage column represents the frequency that the themes were mentioned

Table 7 Research Agenda

Research Calls	%	Research Questions
Lean healthcare implementation process: from planning to sustainability	36%	• How do organisations build foundations for the long-term sustainability of lean in healthcare? • To what extent can the healthcare culture affect lean implementation? • How can healthcare organisations develop a framework for lean implementation and sustainability?
Longitudinal analysis of the results post-implementation in healthcare	34%	• What create or destroy value in the long-term implementation of lean practices in healthcare? • What is the long-term impact of lean implementation in the healthcare culture? • How can lean changes affect patient care in the long term?
Development of standard key performance indicators to assess lean outcomes in healthcare	12%	• To what extent can lean performance indicators be standardized in healthcare? • What are the main key performance indicators that impact lean sustainability in healthcare? • How can performance indicators impact value creation in healthcare?
Comparison of lean implementation in different healthcare areas	11%	• What are the main barriers and enablers to replicating lean into dental services? • Is the healthcare ecosystem suitable for a ‘one fits all strategy’ in lean implementation? • Are there differences in approach when implementing lean in different areas of healthcare (e.g. opened and closed door services)?
Building the lean culture in healthcare organisations	4%	• To what extent could psychological safety help to build a lean culture in the organisation? • How can leadership foster a culture of continuous improvement in healthcare? • To what extent can a lean culture be implemented and sustained in healthcare?
Patient involvement in process improvements	3%	• What is the impact of co-design practices on lean implementation in healthcare? • How can patients engage in process improvements in healthcare? • To what extent can patients create or destroy value in the healthcare ecosystem?

within the literature, and they are presented in order of higher frequency.

The results show six trending themes as research calls in the literature. The first theme relates to understanding lean implementation from the planning to deployment phases. In this theme, we suggest research questions that address the long-term sustainability of lean based on frameworks and elements of its foundation. We also underline the healthcare culture's complexity and how it can affect lean success. One possible research path is to investigate the impact of cultural factors in healthcare; for instance, DeSanctis et al. (2018) underscored some cultural factors (e.g. performance orientation and gender egalitarianism) as successful enablers of the management and sustainability of lean. Concurrently, Erthal and Marques (2018) suggest a research direction for lean sustainability, which would investigate the lean roots in the TPS and try to understand how successful lean organisations in Japan build sustainability.

The second theme reveals two research opportunities still under-researched; the first is using longitudinal studies in lean healthcare. This type of approach to the phenomenon is hardly found in the literature. We suggest that future longitudinal studies can, for example, replicate, confirm, and explore previous cross-sectional studies. This might shed light upon the second under-researched opportunity, the scarcity of studies on the ‘aftermath’ of lean implementation (e.g. Leite et al. 2024). These studies are invaluable to understanding lean failures and sustainability (e.g. Hung et al. 2018). In that vein, Cui and Osborne (2022) suggest that when service does not improve the lives of service users, then value is destroyed, which goes against one of the most germane lean principles. Therefore, our research questions mainly focus on understanding how value can be created or destroyed in the long term in the healthcare ecosystem.

The third theme is about lean performance measurement in healthcare. The literature suggests the need to investigate the development and adoption of standard performance indicators to assess lean outcomes in healthcare. For instance, an SLR conducted by Negrão et al. (2017) identified an adverse effect of lean practices on the operation and financial performance in manufacturing. Using this as an example, the development of standard performance indicators in healthcare can show the causes of such negative effects. Furthermore, a survey with managers across the healthcare sector could identify the main performance indicators that impact lean sustainability. Therefore, our research questions suggest the identification of performance indicators that impact value creation/destruction in healthcare. Furthermore, we urge researchers to investigate how lean performance indicators can be standardised in healthcare.

The fourth research theme touches on the diversity of areas within the healthcare system (e.g. logistics, management, mental health, emergency, general practitioner, and specialized care). As an illustration, the results from our SLR show that only 2% of the lean applications have been conducted in the area of dental care (Fig. 5), representing a research opportunity to expand studies conducted in this area (e.g. Robinson et al. 2016). Another example is the application of lean in public mental healthcare, where Stentoft and Freytag (2020) identified a positive impact of the lean culture on patients' treatment, especially regarding value creation. Therefore, our research questions are oriented toward understanding lean applicability in different healthcare areas, especially those with steady demand (closed doors or by appointment) and variable demand (open doors such as emergency areas and hospitals) (e.g., Leite et al. 2020).

The lean culture and the impact of leadership on supporting lean is the fifth theme that emerged as a research suggestion in the literature. One of the research paths is related to understanding the impact of psychological safety on building a lean culture (e.g. Leite et al. 2024). Regarding this topic, Lindsay, Kumar, and Juleff (2020) suggest that creating psychologically safe spaces and allowing staff to engage in lean initiatives facilitates the breakdown of traditional healthcare hierarchies. Similarly, Tortorella et al. (2021) found an association between organisational culture and leadership styles in companies that adopted lean practices. The study was conducted in manufacturing settings, but in a future study, it could be adapted to understand leadership styles in the complex healthcare environment. Thus, our research questions focus on understanding the culture and leadership impact on lean sustainability.

The last research theme addresses the role of the patient in process improvement. The relevance of patients' involvement in the lean process is underscored by Radnor et al. (2012), who advocate that value is defined from the users' perspective. The importance of the patient's involvement in the improvement was discussed by Al Owad et al. (2018), who used the 'voice of customer' approach (adapted from

Six Sigma) to improve patient flow in the emergency department. Hence, positioning the patient as the value arbiter (Osborne et al. 2022), we suggested research questions that focus on the role of the patient in creating and destroying value in lean healthcare implementation.

To sum up, by identifying specific barriers and the status of lean within healthcare, this study equips practitioners with a deeper understanding of the common barriers to lean adoption and offers strategies to navigate these obstacles. The examples of lean application across different healthcare areas, coupled with an examination of various lean tools and enablers, serve as practical guidance for healthcare managers and lean practitioners. This can help in tailoring lean strategies to specific contexts, enhancing the effectiveness of lean interventions. Conversely, for researchers, the outlined research agenda and questions open avenues for further investigation into critical aspects of lean implementation. By highlighting under-researched areas such as the aftermath of lean implementation and the impact of cultural factors on lean success, the research agenda encourages scholarly exploration that can fill these gaps in the literature.

4.3 Implication to society and policymakers

The findings of the exploration of lean implementation in healthcare have significant implications for society and policymakers, highlighting the transformative potential of lean methodologies to enhance healthcare delivery and patient care. By presenting the barriers and enablers of lean adoption in healthcare, the study provides a roadmap for policymakers aiming to improve efficiency and effectiveness within the health sector. The examples of lean practices across various healthcare settings underscore the adaptability and impact of lean principles, suggesting that wider adoption could lead to substantial improvements in patient outcomes and service delivery.

For society, the emphasis on patient involvement as a cornerstone of lean implementation in healthcare underscores a shift towards more patient-centred care models. This approach not only enhances the quality of care but also empowers patients by placing them at the heart of healthcare processes, ensuring that services are aligned with their needs and values. Conversely, policymakers can draw from our findings to develop regulations and frameworks that support the integration of lean methodologies into healthcare systems. By fostering an environment that encourages lean adoption, policymakers can drive improvements in public and private healthcare services, ultimately benefiting patients and society at large. Moreover, the identification of research opportunities related to lean's long-term sustainability and performance measurement offers guidance for policy formulation that supports continuous improvement and accountability in healthcare services.

5 Concluding remarks

The study sought to identify the factors that impact the implementation and sustainability of lean and value creation in the healthcare ecosystem. Four research questions were derived from the literature and practice to achieve this aim. The RQ1 outlines the current state of the lean literature in healthcare, showing how studies in this area have evolved and spread to different parts of healthcare (Table 2 and Figs. 2, 3, 4, 5, and 6). The RQ2 presents several underlying and meaningful barriers and enablers in the lean journey in healthcare (Tables 3 and 4). The RQ3 offers an overview of the success and failure cases of lean in different areas of healthcare (Tables 5 and 6); a research agenda is derived from the literature (Table 7). Finally, the RQ4 provides invaluable contributions to the literature by presenting a framework that illustrates how value is impacted in the healthcare ecosystem (Fig. 7). This framework discusses the concept of value creation/destruction and the importance of balanced centrality in the healthcare ecosystem.

5.1 Limitations

The study has its limitations. First, this study is drawn on secondary data and a well-known research protocol (e.g., Tranfield et al. 2003) that resulted in a robust database with 134 articles. However, despite the researchers' efforts to select relevant articles for the extraction form, we acknowledge that a small number of papers might be missing from our review. Some of these articles might still be in the reviewing and publication process or were not identified by the search criteria. Therefore, to address this limitation, we suggest future research using a similar protocol with more keywords to enhance the number of articles found.

Second, the presentation of negative and positive results of lean applications presents a limitation (e.g., Narayana-murthy et al. 2018; Rathi et al. 2021). Although our aim was only to exemplify these results, the study only touches the surface of the underlying causes of such results. Thus, we suggest that future studies using different methods (i.e., focus groups, fuzzy Delphi or fuzzy DEMATEL) might provide an in-depth investigation of the underlying causes of positive and negative results of lean applications in healthcare. We acknowledge that a natural step for these and other research findings is to focus on validation from healthcare experts (i.e., managers, policymakers, practitioners, management consultants, and healthcare staff), and therefore, we suggest that future research considers these important stakeholders' perspectives.

Third, as the study does not present empirical data, we suggest using primary data from the healthcare field

as future research. For instance, as suggested in Table 7, longitudinal studies analysing the impact of lean barriers and the use of enablers to counter the inhibitors in the healthcare ecosystem might present a new perspective on lean sustainability. Furthermore, studies that analyse the patient's participation in co-designing new processes in healthcare might shed light on the value creation and destruction concept. Finally, we expect that outcomes from this research will motivate a plethora of theoretical and empirical studies in healthcare operations.

Appendix I enablers: Representative quotes

Themes	Representative Quotes From Literature
Leadership commitment in the implementation	'Leadership was also a key factor in successful projects. The importance of leadership demonstrated itself in two distinct ways. First, the support of organisational leaders was important for the successful implementation of process change' (Laureani et al. 2013, p.333) 'The workforce flexibility, transformational leadership and team leadership foster the change to sustainable lean health care' (Abdallah and Alkhaldi 2014, p.501)
Lean as an organisational strategy	'Integrating the six-sigma culture into entire organizations by the commitment and involvement of top management can multiply the positive effects and make a significant impact at all levels' (Antony et al. 2007, p.339) 'The next step for the case organisation might be to create a common organisational culture that is supported by continuous improvements, focusing on value-adding activities for the residents and Other stakeholders. By doing so, the organisation will have a clear vision and philosophy, which makes it less vulnerable with time when certain key people no longer remain in the organisation' (Andersson and Hiltefth 2015, p.69)

Themes	Representative Quotes From Literature	Themes	Representative Quotes From Literature
Team involvement	‘High improvement clinics reported higher baseline levels of teamwork, workforce engagement in the form of personal recognition for ideas to improve care, and staff efficacy to implement change’ (Hung et al. 2019, p. 73) ‘Over the course of the second phase of lean implementation an important enabler was the increase in the level of staff engagement in the improvement programme. Here we refer to staff engagement as a collective, team-level experience’ (Radcliffe et al. 2020, p.875)	Insufficient preparation for implementation	‘Our research leads us to conclude that Oman’s government health-care sector exhibits sufficient characteristics of the required readiness factors for piloting lean projects within hospital departments’ (Sohal et al. 2021, p.21) ‘We can conclude that no framework or assessment tool for the lean readiness of healthcare institutions exist’ (Narayanamurthy et al. 2018, p.125)
Planning for Lean implementation	‘Factors like leadership, culture of organizations, communication of goals and proper training are important for lean application’ (Ajmera and Vineet 2019, p. 379) ‘Plan to initiate the change’ (Morell-Santandreu et al. 2021, p.7)	Resistance to change	‘Scepticism and resistance will be high, success not guaranteed. Organizations may think twice before embracing on such a journey, or worse, superficially implement lean thinking, adding to existing resistance and application of lean thinking to health care making it more difficult to improve health care in the long term’ (Joosten et al. 2009, p.345) ‘The top 5 limitations of Lean in the healthcare services are Lean is new in the healthcare sector that makes it difficult to implement, lack of empirical evidence to convince top management, lack of coordination between functions, managerial resistance to change and employee resistance to change’. (Antony et al. 2019, p.1380)

Appendix II barriers: Representative quotes

Themes	Representative Quotes From Literature	Themes	Representative Quotes From Literature
Inadequate understanding of Lean thinking	‘Although many interview participants indicated enthusiasm for embedding lean culture on an organizational level across the hospital, extending lean implementation across other specialties may be of limited success if the healthcare context does not align at all with the manufacturing origins of lean, associated with the standardisation of processes and the minimisation of variance’ (Radcliffe et al. 2020, p.882) ‘The penetration of lean into healthcare and the results followed by implementation are somewhat confusing’ (Turki 2016, p.15)	Lack of leadership	‘Healthcare leaders are usually skilled in their own fields but are not experts in people management and They have often limited knowledge about how to motivate, coach, and engage employees. Healthcare leaders frequently work in clinical practice and Only manage healthcare units part-time. They have often limited opportunities and interest in acting as facilitators and coaches to healthcare staff. Professional knowledge is also a barrier in this respect.’ (Poksinska 2014, p.191) ‘Impeding factors to the implementation of six sigma in healthcare organizations areas follows: lack of financial resources; lack of human resources; lack of time; lack of leadership.’ (Antony et al. 2007, p.339)

Themes	Representative Quotes From Literature
Poor work processes	‘While investments are needed to build and develop medical infrastructures in many countries, their impact will remain limited if operational and structural inefficiencies continue to plague the daily operations of health systems.’ (Mahmoud et al. 2021, p.637) ‘Thus, team members were able to identify different activities that did not add value for the patient or which caused rework for themselves, long waits, unnecessary movements, or repeated errors.’ (Prado-Prado et al. 2020, p.15)

Authors’ contribution Higor Leite: Conceptualization, Methodology; Data curation, Writing- Original and final draft preparation. Flávia Quadros: Literature Review, Validation, Writing- Reviewing and Editing.

Funding No funds, grants, or other support was received.

Data availability Not applicable.

Declarations No potential conflict of interest is reported by the authors. No funds, grants, or other support was received.

Ethics approval Not applicable.

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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